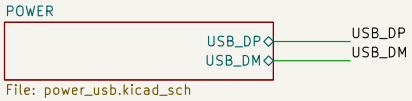
Each sheet instance adds an offset to its references: POWER 10, MCU 30, CH1 100, CH2 200.

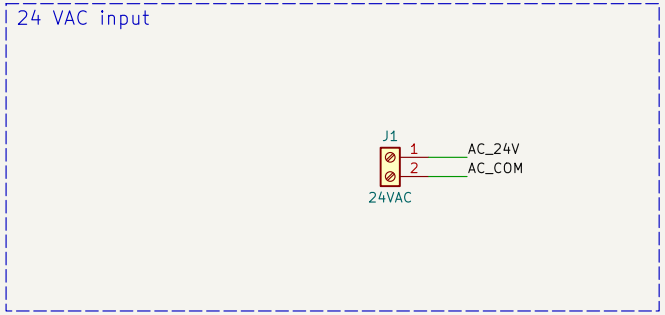
CH1 and CH2 are two instances of one file, ac_channel.kicad_sch.

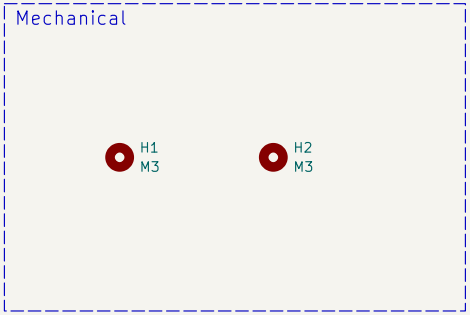












ZoneFlow hardware toolchain test

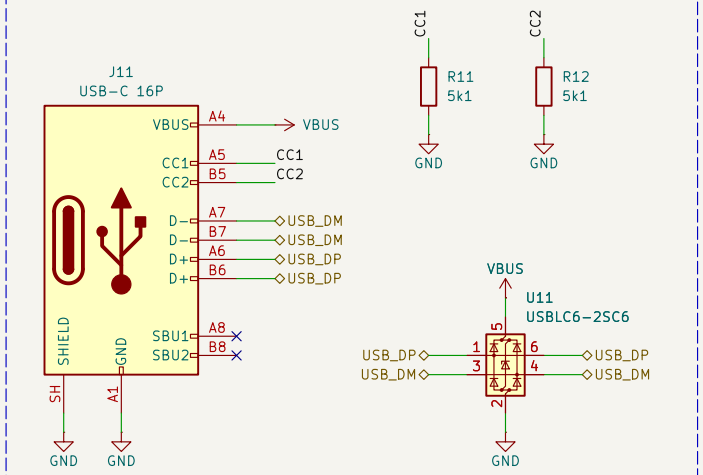
ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /
File: toolchain_test.kicad_sch

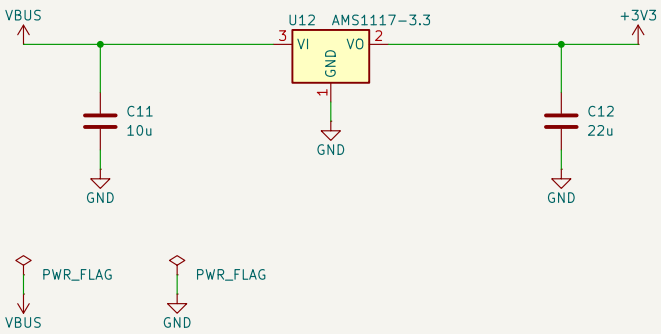
Title: ZoneFlow hardware toolchain test

Size: A4	Date: 2026-11-10	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 1/5

USB-C receptacle, CC pull-downs, ESD



3.3 V regulator



USB powers the logic rail only.

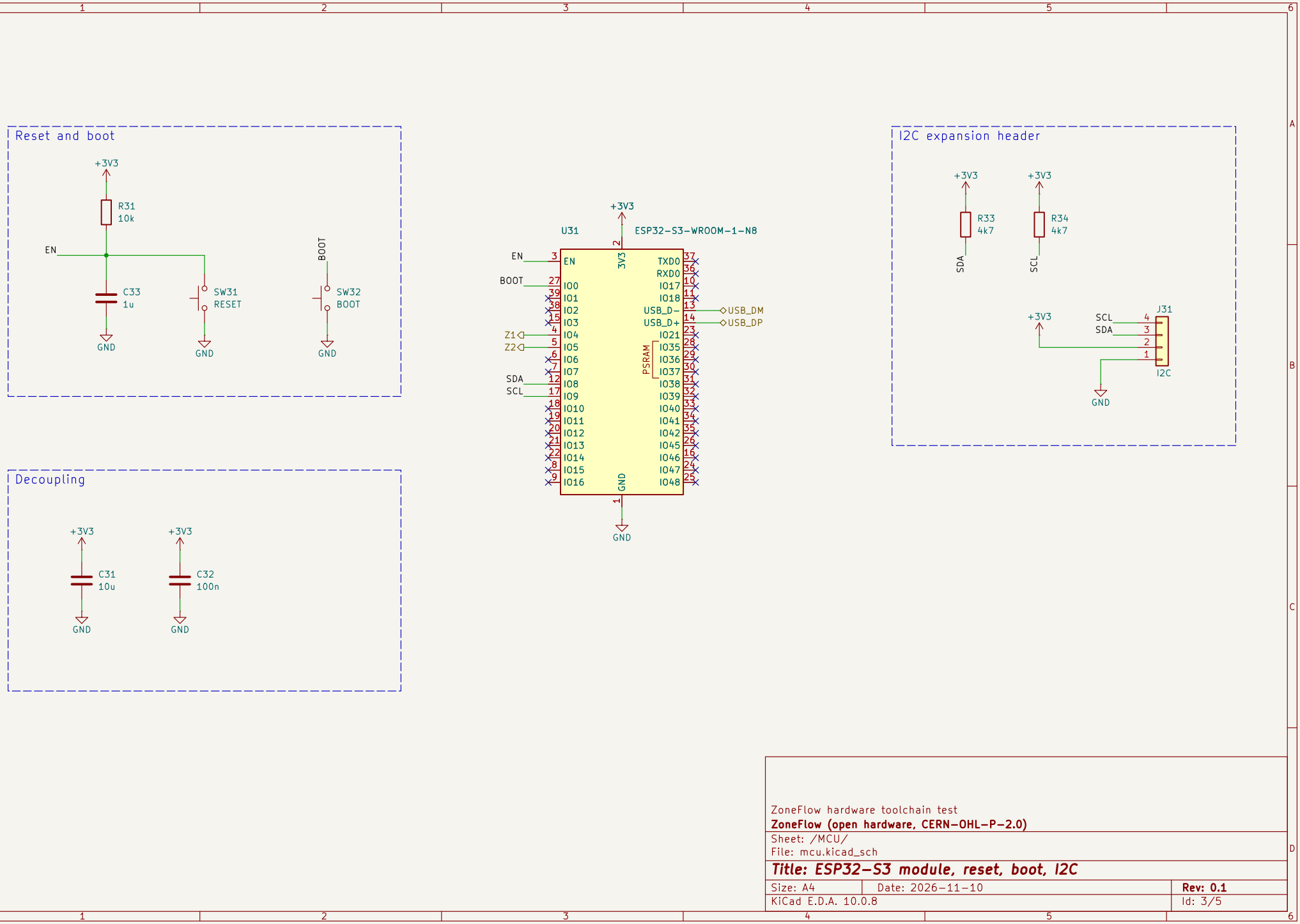
ZoneFlow hardware toolchain test

ZoneFlow (open hardware, CERN-OHL-P-2.0)

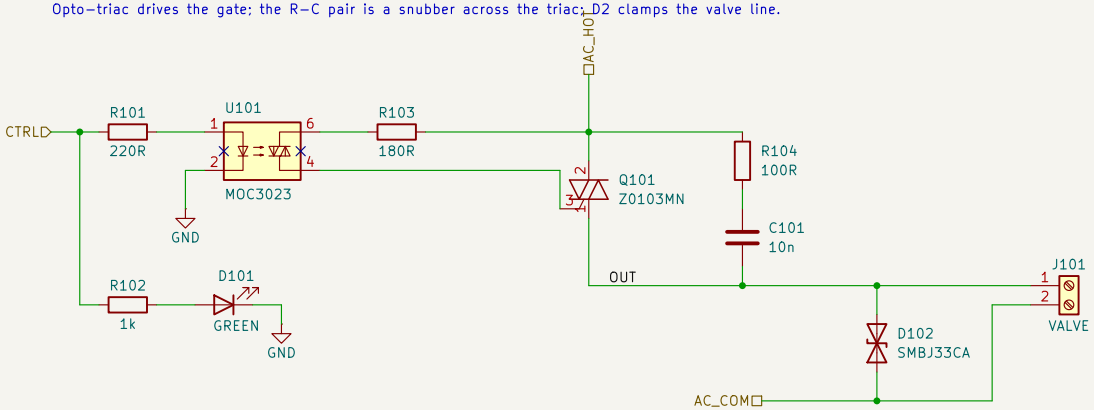
Sheet: /POWER/
File: power_usb.kicad_sch

Title: USB-C input and 3.3 V regulator

Size: A4	Date: 2026-11-10	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 2/5


Reset and boot
+3V3
R31
10k
EN
C33
1u
GND
SW31
RESET
GND
BOOT
SW32
BOOT
GND
Decoupling
+3V3
C31
10u
GND
+3V3
C32
100n
GND
U31
ESP32-S3-WROOM-1-N8
+3V3
EN
BOOT
Z1
Z2
SDA
SCL
3V3
GND
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
PSRAM
USB_DM
USB_DP
GND
I2C expansion header
+3V3
R33
4k7
SDA
+3V3
R34
4k7
SCL
+3V3
SCL
SDA
J31
I2C
GND
ZoneFlow hardware toolchain test
ZoneFlow (open hardware, CERN-OHL-P-2.0)
Sheet: /MCU/
File: mcu.kicad_sch
Title: ESP32-S3 module, reset, boot, I2C
Size: A4
Date: 2026-11-10
Rev: 0.1
KiCad E.D.A. 10.0.8
Id: 3/5

Opto-triac drives the gate; the R-C pair is a snubber across the triac; D2 clamps the valve line.



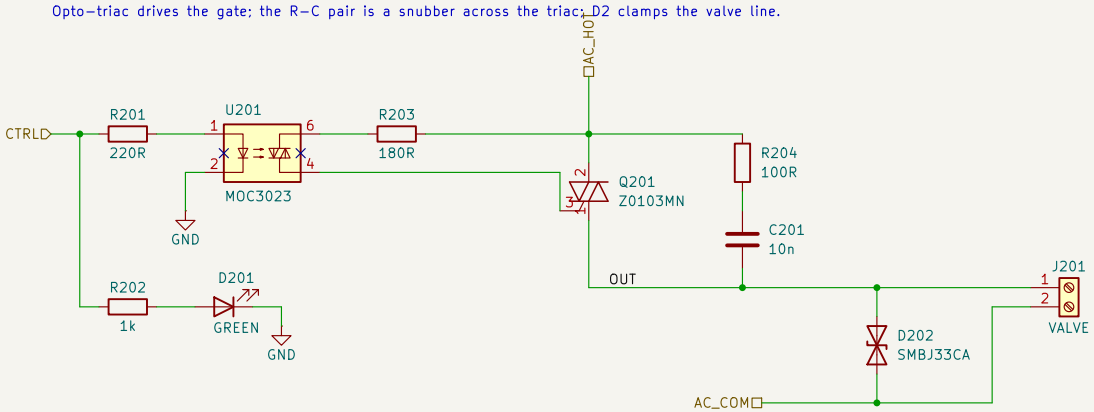
ZoneFlow hardware toolchain test

ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /CH1/
File: ac_channel.kicad_sch

Title: Isolated 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 4/5



ZoneFlow hardware toolchain test

ZoneFlow (open hardware, CERN-OHL-P-2.0)

Sheet: /CH2/
File: ac_channel.kicad_sch

Title: Isolated 24 VAC output channel

Size: A4	Date: 2026-11-10	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 5/5